

Information Security

Security Controls for CIA

- Now move from:
- **"What went wrong?"**
- to:
- **"How do we prevent it?"**



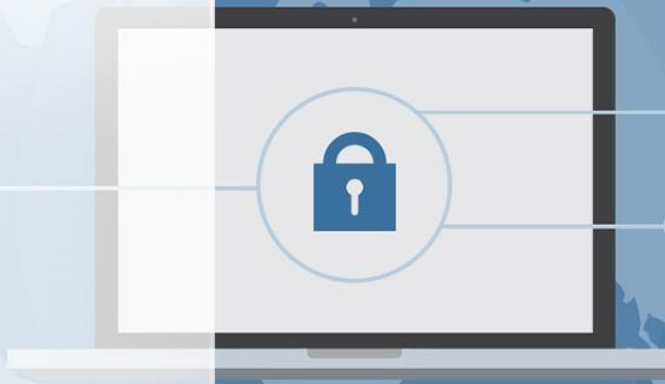
CIA Controls



Availability Controls

Examples:

- Backups
- Redundancy
- Load balancing
- Failover systems
- Disaster recovery
- DDoS protection
- UPS
- High-availability architecture
- Incident response



Backups

- A backup is a copy of important data stored separately so it can be restored if the original data is lost, damaged, or deleted.

  Main objective is Data recovery after deletion, hardware failure, ransomware, or other incidents.



Redundancy

Redundancy means having extra components or systems available to take over when the primary component fails. For example A server has two power supplies instead of one.

It Prevents a single component failure from stopping the service.



Load Balancing

- Load balancing distributes network traffic or workload across multiple servers.
- Like 10,000 users accessing a website are distributed among several web servers.
- It Prevents one server from becoming overloaded and improves availability.



Failover Systems

A failover system automatically switches from a failed primary system to a backup system. If Server A fails, Server B automatically starts serving users. It Minimize service interruption.



Disaster Recovery (DR)

Disaster recovery is a planned process for restoring IT systems, applications, and data after a major disruption. Such as After a data center fire, a company restores its services using a backup data center. It Resume critical operations as quickly as possible.



DDoS Protection

DDoS protection detects and mitigates Distributed Denial-of-Service attacks that attempt to make a service unavailable by sending huge amounts of traffic. Likewise, A web security service filters malicious traffic before it reaches the company's server.

Objective is to Keep websites and online services available during attacks.



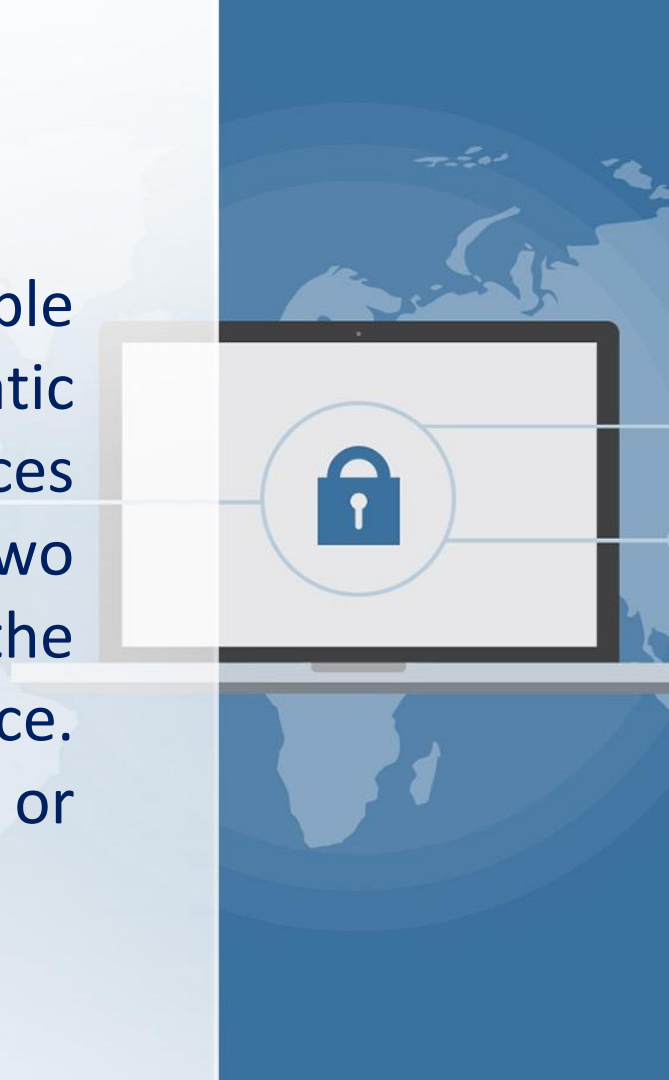
UPS (Uninterruptible Power Supply)

A UPS provides temporary electrical power when the main power supply fails. A server remains powered for several minutes during a power outage. It Prevent sudden shutdowns and provide time for safe shutdown or backup power activation.



High-Availability (HA) Architecture

High-availability architecture uses multiple systems, redundancy, and automatic recovery mechanisms to keep services running with minimal downtime. Two servers run together, and if one fails, the other continues providing the service. Purpose is to Achieve continuous or near-continuous service availability.



Incident Response

Incident response is a structured process for detecting, handling, containing, and recovering from security incidents. Like During a ransomware attack, the security team isolates infected computers and restores affected systems from clean backups.

It Reduce damage, recovery time, and service disruption.



Incident Response

Detection

↓

Analysis

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Containment

↓

Eradication

↓

Recovery

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Lessons Learned



Comparison

Control	Main Question It Answers
Backup	How do we recover lost data?
Redundancy	What happens if one component fails?
Load Balancing	How do we prevent overload?
Failover	How do we continue service after failure?
Disaster Recovery	How do we restore operations after a major disaster?
DDoS Protection	How do we keep services available during traffic attacks?
UPS	How do we handle temporary power failure?
High Availability	How do we minimize overall downtime?
Incident Response	How do we respond when an incident disrupts services?



The Core Mental Model

Confidentiality = Don't let the wrong people see it.

Integrity = Don't let the wrong people change it.

Availability = Don't let the wrong circumstances prevent people from using it.



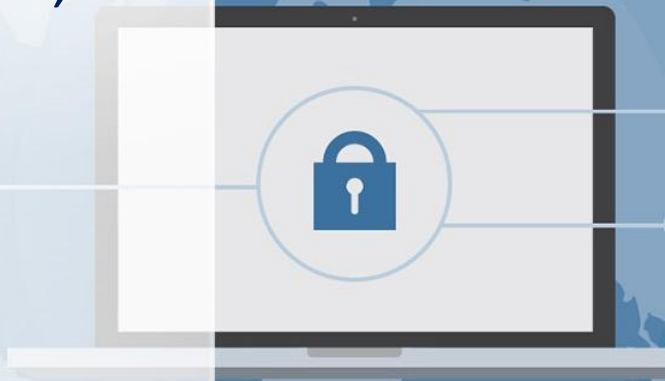
Final Takeaway

When analyzing any cybersecurity incident, don't immediately ask "What attack is this?"

First ask: **What happened to the information or service?**

Then determine:

Was it exposed? → C
Was it changed? → I
Was it unavailable? → A



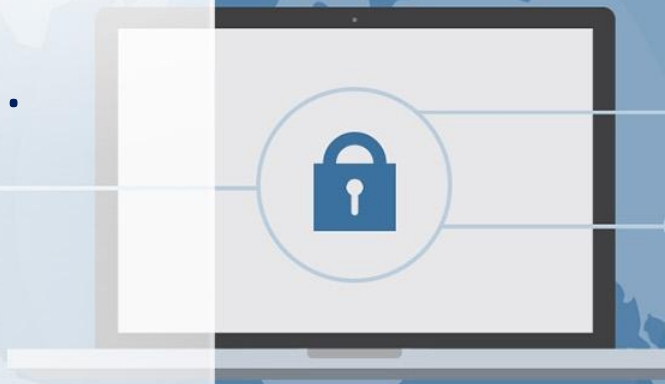
Counter Measures to keep your system and data safe

- Back up your data
- Use strong passwords
- Take care when working remotely
- Be wary of suspicious emails
- Install anti-virus and malware protection
- Make sure your Wi-Fi is secure
- Keep software up-to-date
- Secure personal info offline
- Use Two-Factor Authentication



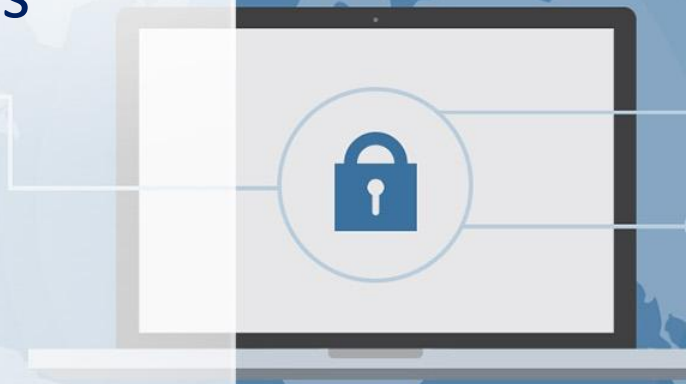
Discussion
Group 1 — Social Media

- An attacker takes over a user's account and posts fake information.
- **Determine:**
- CIA impact
- Attack
- Security control



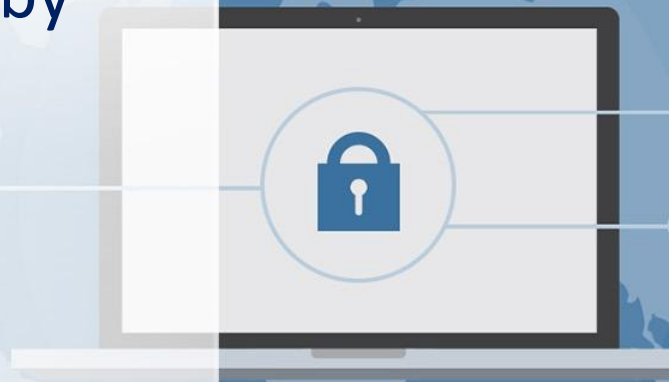
Group 2 — University

- An unauthorized person downloads student examination results.
- **Determine:**
- CIA impact
- Vulnerability
- Control



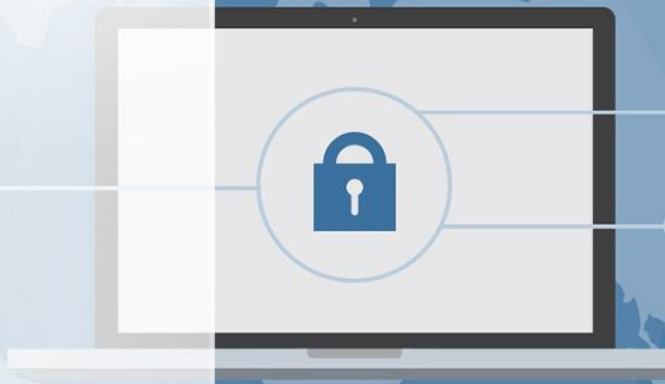
Group 3 — Banking

- A transaction database is modified by an unauthorized employee.
- **Determine:**
- CIA impact
- Threat
- Control



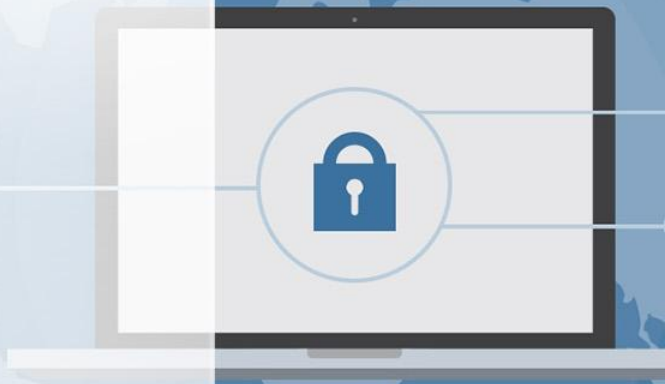
Group 4 — E-Commerce

- Attackers overload an online shopping website during a major sale.
- **Determine:**
- CIA impact
- Attack
- Control



Group 5 — Hospital

- Ransomware encrypts patient records.
- **Determine:**
- CIA impact
- Possible consequences
- Controls



Availability Controls Task

A university online exam system faces **server overload, server failure, power outage, DDoS attack, data loss, and a data-center fire**.

Task: Match each problem with the correct availability control:

Server overload → _____

Server failure → _____

Power outage → _____

DDoS attack → _____

Data loss → _____

Data-center fire → _____

Cyberattack handling → _____

Overall minimum downtime → _____

Extra backup components → _____

